GEOMETRY · UNIT 2: ANGLES AND PARALLEL LINES

Parking Lot Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design the stripes for a school parking lot. They identify angle pairs formed by parallel lines and a transversal, use algebra to find angle measures, prove lines parallel, use triangle angle sums for traffic islands, and design a lot that meets the rules.

Parallel lines cut by transversals appear in every parking lot. Students name angle pairs, use algebra, justify with theorems, and then design within constraints.

Standards Addressed

- G-CO.9 Prove theorems about lines and angles
- G-CO.10 Prove theorems about triangles (angle sum)
- G-CO.12 Make formal geometric constructions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with an aerial photo. Part 1 angle pairs.
Day 2	Part 2 algebra, proving lines parallel, triangle islands.
Day 3	Part 3 design the lot.

Materials

- Student handout
- Protractors and rulers
- Graph or grid paper for the lot design
- Optional: aerial photo of the school lot

How the Three Versions Differ

Level	What changes
CP2	Name angle pairs and find measures; one-variable equations with a guide.
CP1	Multi-step angle chases, converses to prove lines parallel, and exterior angles.
Honors	Systems of equations from angle relationships, two-column proofs, and a design with a parallel-line proof.

Watch For (Common Misconceptions)

- Thinking same-side interior angles are congruent.
- Using angle-pair theorems when lines are not parallel.
- Mixing up a theorem and its converse.

Extension Ideas

- Measure angles in a real lot.
- Compare angled and straight parking capacity.
- Construct parallel lines with a compass.

Answer Key – CP2

Part 1: Angle Pairs on the Lot

#	Task	Answer
1	Name a pair of corresponding angles, alternate interior angles, and same-side interior angles in the diagram.	Examples: ∠1 and ∠5 corresponding; ∠2 and ∠7 alternate interior; ∠2 and ∠8 same-side interior.
2	If m∠2 = 60°, find the measures of the other seven angles.	∠3, ∠6, ∠7 = 60°; ∠1, ∠4, ∠5, ∠8 = 120°
3	Which angle pairs are congruent and which are supplementary?	Corresponding, alternate interior, alternate exterior, and vertical angles are congruent; same-side interior angles and linear pairs are supplementary.

Part 2: Use Algebra

#	Task	Answer
4	Alternate interior angles measure (3x + 15)° and (5x − 25)°. Find x and the angle measure.	3x + 15 = 5x − 25 → x = 20; 75°
5	Same-side interior angles measure (2y + 10)° and (4y − 10)°. Find y and both angles.	6y = 180 → y = 30; 70° and 110°
6	A triangular traffic island has angles x°, 2x°, and 3x°. Find each angle.	6x = 180 → x = 30; 30°, 60°, 90°
7	An exterior angle of the island measures 130°. One remote interior angle is 70°. Find the other.	130 − 70 = 60°

Part 3: Design Your Lot

#	Task	Answer
8	Draw a curb and an aisle edge that are parallel. Choose an angle for your spaces and draw at least five stripes.	Answers vary.
9	Label all eight angles at one stripe and explain how you found them.	Answers vary.
10	Add a triangular island and label its angles.	Answers vary.

Scoring note: Angle labels: at each intersection 1 (or 5) is upper left, 2 (or 6) upper right, 3 (or 7) lower left, 4 (or 8) lower right. In the diagram the acute angles are 2, 3, 6 and 7.

Answer Key – CP1

Part 1: Angle Pairs on the Lot

#	Task	Answer
1	If m∠2 = (4x + 8)° and m∠7 = (6x − 12)°, what is the relationship between ∠2 and ∠7? Find x and every angle.	Alternate interior angles are congruent: 4x + 8 = 6x − 12 → x = 10. ∠2, ∠3, ∠6, ∠7 = 48°; ∠1, ∠4, ∠5, ∠8 = 132°.
2	If m∠1 = 2a° and m∠7 = (a + 30)°, find a and both angles. Which theorem did you use?	Same-side interior angles are supplementary: 2a + a + 30 = 180 → a = 50; ∠1 = 100°, ∠7 = 80°.
3	Explain why ∠3 and ∠6 are congruent using a chain of two theorems.	∠3 ≅ ∠2 (vertical angles) and ∠2 ≅ ∠6 (corresponding angles), so ∠3 ≅ ∠6 (transitive property).

Part 2: Is It Really Parallel?

#	Task	Answer
4	Two stripes cross the curb at 62° and 62° as corresponding angles. Are the stripes parallel? Which converse proves it?	Yes, by the Converse of the Corresponding Angles Postulate.
5	Same-side interior angles between two stripes and the curb measure 117° and 65°. Are the stripes parallel? Explain.	No: 117 + 65 = 182 ≠ 180.
6	A triangular island has angles (x + 10)°, (2x − 5)°, and (3x + 25)°. Find all three angles and the exterior angle at the largest angle.	6x + 30 = 180 → x = 25: 35°, 45°, 100°; exterior angle at the 100° angle is 80°.
7	Stripe angles must be between 45° and 75°. If a stripe makes an angle of (3k + 9)° with the curb, write and solve a compound inequality for k.	45 ≤ 3k + 9 ≤ 75 → 36 ≤ 3k ≤ 66 → 12 ≤ k ≤ 22

Part 3: Design Your Lot

#	Task	Answer
8	Draw your design and label one stripe's eight angles.	Answers vary.
9	Show, using a converse, that your two rows of stripes are parallel to each other.	Answers vary.
10	Add an island and label its interior and exterior angles.	Answers vary.

Answer Key – Honors

Part 1: Systems from Angles

#	Task	Answer
1	$m\angle 1 = (3x + y)°$, $m\angle 5 = 110°$, $m\angle 2 = (2x - y)°$, and $m\angle 6 = 70°$. Write a system and solve for x and y.	$\angle 1 \cong \angle 5$ and $\angle 2 \cong \angle 6$ (corresponding): $3x + y = 110$ and $2x - y = 70$. Adding: $5x = 180 \rightarrow x = 36$, $y = 2$.
2	$m\angle 3 = (x^2 - 5x)°$ and $m\angle 6 = (2x + 18)°$. Find all possible values of x that make the angles valid.	$\angle 3$ and $\angle 6$ are alternate exterior, so $x^2 - 5x = 2x + 18 \rightarrow x^2 - 7x - 18 = 0 \rightarrow x = 9$ or $x = -2$. $x = 9$: 36°; $x = -2$: 14°. Both give positive angles, so both are valid.
3	Prove: If $m \parallel n$, then $\angle 1$ and $\angle 7$ are supplementary.	1. $m \parallel n$ (given). 2. $\angle 1 \cong \angle 5$ (Corresponding Angles Postulate). 3. $\angle 5$ and $\angle 7$ form a linear pair, so $m\angle 5 + m\angle 7 = 180°$ (Linear Pair Postulate). 4. $m\angle 1 + m\angle 7 = 180°$ (substitution). 5. $\angle 1$ and $\angle 7$ are supplementary (definition).

Part 2: Proofs & Islands

#	Task	Answer
4	Prove the Triangle Angle Sum Theorem using a line through one vertex parallel to the opposite side.	Draw a line through the top vertex parallel to the base. Alternate interior angles make the three angles of the triangle fit along the straight line, so they add to 180°.
5	A triangular island has an exterior angle of $(5x - 6)°$ and remote interior angles of $(2x + 8)°$ and $(x + 22)°$. Find all interior angles.	$5x - 6 = 3x + 30 \rightarrow x = 18$: angles 44°, 40°, and $180 - 84 = 96°$
6	Two stripes make angles of $(4k - 3)°$ and $(2k + 39)°$ with the curb at corresponding positions. For what k are they parallel? Is that k allowed by…	$4k - 3 = 2k + 39 \rightarrow k = 21$: angle 81°. Not allowed (more than 75°).

Part 3: Design & Prove

#	Task	Answer
7	Draw the design and give every angle in terms of θ.	Answers vary.
8	Write a proof that your crosswalk makes equal angles with every stripe.	Answers vary.
9	Choose θ so that the island's angles are θ, $2\theta - 30$, and a third angle that is a whole number between 45 and 75. Show your work.	$\theta + (2\theta - 30) + \text{third} = 180 \rightarrow \text{third} = 210 - 3\theta$. $45 \le 210 - 3\theta \le 75 \rightarrow 45 \le \theta \le 55$. Example $\theta = 50$: angles 50°, 70°, 60°.

Scoring note: Angle labels: at each intersection 1 (or 5) is upper left, 2 (or 6) upper right, 3 (or 7) lower left, 4 (or 8) lower right. In the diagram the acute angles are 2, 3, 6 and 7.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Design follows the rules (parallel stripes, angles between 45° and 75°) with every angle justified. Goes beyond the minimum with a creative, realistic choice.	Design follows the rules (parallel stripes, angles between 45° and 75°) with every angle justified.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Blueprint shows parallel stripes, labeled angle measures with reasons, and a traffic island. Polished, organized, and visually engaging.	Blueprint shows parallel stripes, labeled angle measures with reasons, and a traffic island. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.